Topic 2.10 - Inverses of Exponential Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

Logarithmic functions arise by reversing exponential functions. The horizontal asymptote of the exponential becomes a vertical asymptote of the logarithm, and the exponential range becomes the logarithm domain. A transformed inverse must preserve every shift, scale, and restriction.

Learning targets

- Explain logarithms as inverse functions of exponentials.
- Connect inverse pairs through compositions, points, graphs, domains, ranges, and asymptotes.
- Construct inverses of basic and transformed exponential functions.

Language and structure

Term	Meaning in this topic
inverse pair	Two functions whose compositions return the original input on appropriate domains.
exponential-log identity	$\log_b(b^x) = x$ and $b^{\log_b x} = x$ for $x > 0$.
vertical asymptote	A vertical line approached by a logarithmic graph near the edge of its domain.
domain-range swap	The domain of one inverse partner equals the range of the other.
reflection	The graphs of inverse functions are reflections across $y = x$.

Launch: make a claim before calculating

Launch investigation

Reasoning first

The function $f(x) = 2^x$ maps 3 to 8. What must its inverse do with 8? Use this fact to explain the meaning of $\log_2 8$.

Concept 1: Exponential and logarithmic identities

Core idea

For $b > 0$, $b \neq 1$, the inverse of $f(x) = b^x$ is $f^{-1}(x) = \log_b x$.

- $\log_b(b^x) = x$ for every real x.
- $b^{\log_b x} = x$ only for $x > 0$.
- These identities are composition statements, not product rules.

Worked example - annotate the reasoning

Problem. Simplify $\log_5(5^{2x-3})$ and $5^{\log_5(4x+1)}$, stating any restriction.
Model reasoning. The first is $2x - 3$ for all real x. The second is $4x + 1$ provided $4x + 1 > 0$, so $x > -1/4$.

Check your understanding

Independent

Simplify $\ln(e^{x^2+2})$ and $e^{\ln(7-x)}$.

Concept 2: Points, graphs, and asymptotes

Core idea

If (a, b) lies on an exponential function, then (b, a) lies on its logarithmic inverse.

- The exponential domain $\mathbb{R}$ becomes the logarithmic range $\mathbb{R}$.
- The exponential range (k, ∞) or $(-\infty, k)$ becomes the inverse domain.
- The horizontal asymptote $y = k$ reflects to the vertical asymptote $x = k$.

Worked example - annotate the reasoning

Problem. For $f(x) = 2^x + 3$, state the inverse, its domain and range, and its vertical asymptote.
Model reasoning. $f^{-1}(x) = \log_2(x - 3)$. Domain $(3, \infty)$, range $\mathbb{R}$, and vertical asymptote $x = 3$.

Check your understanding

Independent

If $(4, 85)$ lies on $g(x) = 5(2)^x + 5$, identify the corresponding inverse point.

Concept 3: Inverse of a transformed exponential

Core idea

Isolate the exponential expression before applying a logarithm.

- For $f(x) = ab^{x-h} + k$, solve $\dfrac{y-k}{a} = b^{x-h}$.
- The sign of a determines which side of k forms the range and inverse domain.
- Check the inverse by composition or by reversing a known point.

Worked example - annotate the reasoning

Problem. Find the inverse of $f(x) = 3(2)^{x-1} + 4$.
Model reasoning. $y - 4 = 3(2)^{x-1}$, so $(y-4)/3 = 2^{x-1}$ and $x = 1 + \log_2((y-4)/3)$. Thus $f^{-1}(x) = 1 + \log_2((x-4)/3)$, domain $x > 4$.

Check your understanding

Independent

Find the inverse of $g(x) = 5 - 2(3)^x$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Find the inverse of $f(x) = 4^x - 6$.

Exit ticket 2

No calculator

State the vertical asymptote of the inverse of $g(x) = 7(0.5)^x + 2$.

Exit ticket 3

No calculator

Simplify $3^{\log_3(x-1)}$ and state the domain.